Sardar Vallabhbhai National Institute of Technology, Surat
Department of Computer Science and Engineering
M.Tech(CSE/DS) - I Sem - 2026-27
Computer Vision and Image Processing (CSCS111/CSDS119)
Lab Assignment 8 - Motion Analysis

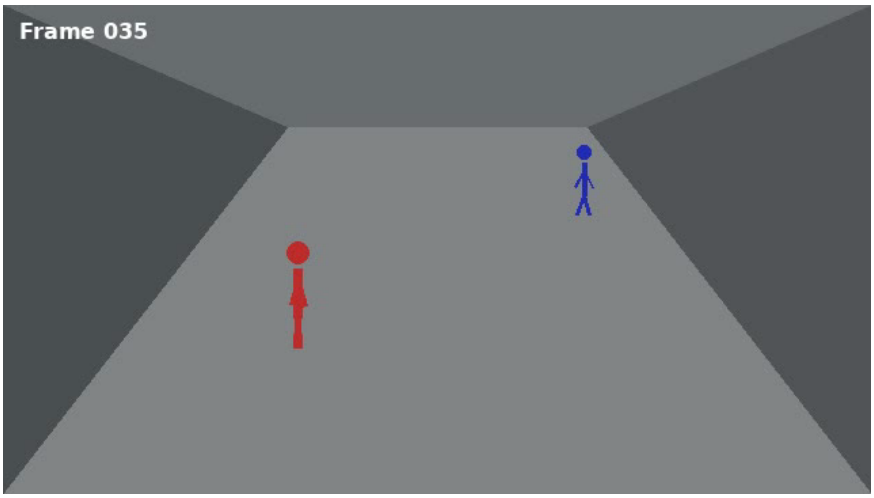
Input: `motion_corridor.avi`
Problem: A corridor surveillance camera records two people moving through a scene. Analyse the video to identify moving regions and investigate when motion detection can fail.

1. Load the provided video and display selected frames. Convert frames to grayscale and observe changes between consecutive frames.

2. Implement frame differencing and background subtraction using MOG2. Display the motion masks and compare how threshold settings affect detected regions.

3. Compare both methods on a segment where one person stops or illumination changes. Record differences in detected motion.

Challenge: Can a motion detector distinguish a stationary person from the background? Explain one limitation using evidence from the video and suggest one improvement.
- Generate Sample frames, frame-difference mask, background-subtraction mask, and a short comparison. Optical flow is optional.
- Use cv2.VideoCapture and cv2.createBackgroundSubtractorMOG2().



Sample frame from the provided video